NGOC LUU

AI Researcher | Efficient AI, Reasoning, and Natural Language Processing

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RESEARCH PROFILE

AI researcher working on efficient learning and reasoning in foundation models. My work spans state-based fine-tuning, test-time scaling, rehearsal-free continual learning, machine unlearning, and low-resource natural language processing. I am currently an AI Researcher at Qualcomm AI Research under the guidance of Dr. Tung Pham. I am seeking Computer Science PhD positions for Fall 2027.

RESEARCH INTERESTS

Efficient adaptation and inference · LLM reasoning and test-time scaling · Continual learning and unlearning · Natural language processing

EDUCATION

Hanoi University of Science and Technology *2020 – 2024*

Bachelor of Computer Science GPA: 3.62/4.00

Undergraduate thesis: *Continual Learning for Relation Extraction*

RESEARCH EXPERIENCE

Qualcomm AI Research Oct 2025 – Present

AI Researcher *Vietnam*

- Conduct research on memory- and compute-efficient adaptation and inference-time reasoning for transformer models and LLMs with Dr. Tung Pham.
- Co-developed Mixture-of-Control, a state-aware fine-tuning framework that enables cross-block information exchange while preserving memory and computational efficiency.
- Developed stochastic backtracking methods for test-time scaling that revisit historical reasoning states to improve the accuracy–token trade-off.

Data Science Laboratory, SoICT Oct 2022 – May 2024

Undergraduate Research Assistant *Vietnam*

- Researched rehearsal-free continual relation extraction with Dr. Linh Ngo, focusing on catastrophic forgetting, prompt selection, and knowledge consolidation without retaining prior-task data.
- Developed a task-specific prompt-pool approach that captures within-task variation while improving separation across tasks.
- Completed an undergraduate thesis that led to an AAAI 2025 oral publication.

SoICT, Hanoi University of Science and Technology Oct 2023 – Mar 2024

Undergraduate Research Assistant *Vietnam*

- Conducted research on Vietnamese legal question answering under Prof. Le Thanh Huong.
- Contributed to the construction of Vietnamese legal provisions and question-answer datasets for training and evaluation.
- Built a retrieval-augmented system and fine-tuned compact language models for answer generation in a low-resource legal domain; the work was published in ACM TALLIP.

PUBLICATIONS AND MANUSCRIPTS

Mixture-of-Control: State-Aware Fine-Tuning for Transformer-based Models *NeurIPS 2026*
Duc Anh Nguyen*, **Tien Ngoc Luu***, Tung Pham, Toan Tran

Beyond the Frontier: Stochastic Backtracking for Efficient Test-Time Scaling *ICML 2026 Workshop*
Dao Tran, Duc Anh Le, **Ngoc Luu**, Quan Pham, Tung Pham, Hung Bui

Adaptive Prompting for Continual Relation Extraction: A Within-Task Variance Perspective *AAAI 2025 – Oral*
Minh Le*, **Tien Ngoc Luu***, An Nguyen*, Thanh-Thien Le, Trang Nguyen, Tung Thanh Nguyen, Linh Ngo, Thien Huu Nguyen

Forget the Fact, Keep the Form: Distribution Redistribution for Retain-Free MDLM Unlearning *Submitted to ICLR 2027*
Ngoc Luu*, Giap Nguyen*, Dinh Phung, Tung Pham

Optimizing Answer Generator in Vietnamese Legal Question Answering Systems Using Language Models *ACM TALLIP 2025*
Huong Le, **Ngoc Luu**, Thanh Nguyen, Tuan Dao, Sang Dinh

*Equal contribution.

ENGINEERING EXPERIENCE

Prep Oct 2024 – Oct 2025
AI Engineer *Vietnam*

· Developed production systems for automated IELTS Speaking and Writing assessment.
· Built speech and text evaluation models for criterion-level scoring and feedback generation.

CMC ATI Mar 2024 – Oct 2024
AI Engineer *Vietnam*

· Developed an internal retrieval-augmented chatbot and a named-entity recognition model for extracting information from official documents.
· Improved automatic speech recognition models for Vietnamese speech-to-text applications.

TECHNICAL SKILLS

Programming	Python, C++
Machine Learning	PyTorch, Transformers, large language models, retrieval-augmented generation, fine-tuning
Research Areas	Efficient adaptation, test-time scaling, continual learning, machine unlearning, relation extraction
Mathematics	Linear algebra, numerical optimization, probability, statistics
Languages	Vietnamese (native), English (fluent)

HONORS AND ACHIEVEMENTS

- Top 2, Spoken Language Understanding track, SoICT Hackathon, 2024.
- Awarded two Academic Merit Scholarships at Hanoi University of Science and Technology.
- Top 0.01% in the Vietnam National High School Graduation Examination, 2020.